Composable Architecture for AI Systems: APIs, MCP, Skills, Sub-Agents, and Workflows

Nordic APIs Summit 2026

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OpenAPI Initiative (OAI)

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As Little AI As Possible

- The line between AI and deterministic code keeps moving
- Good architecture is what lets you move it
- All of it rests on foundations this talk mostly takes for granted

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A Trillion Operations for 16'968 ÷ 404?

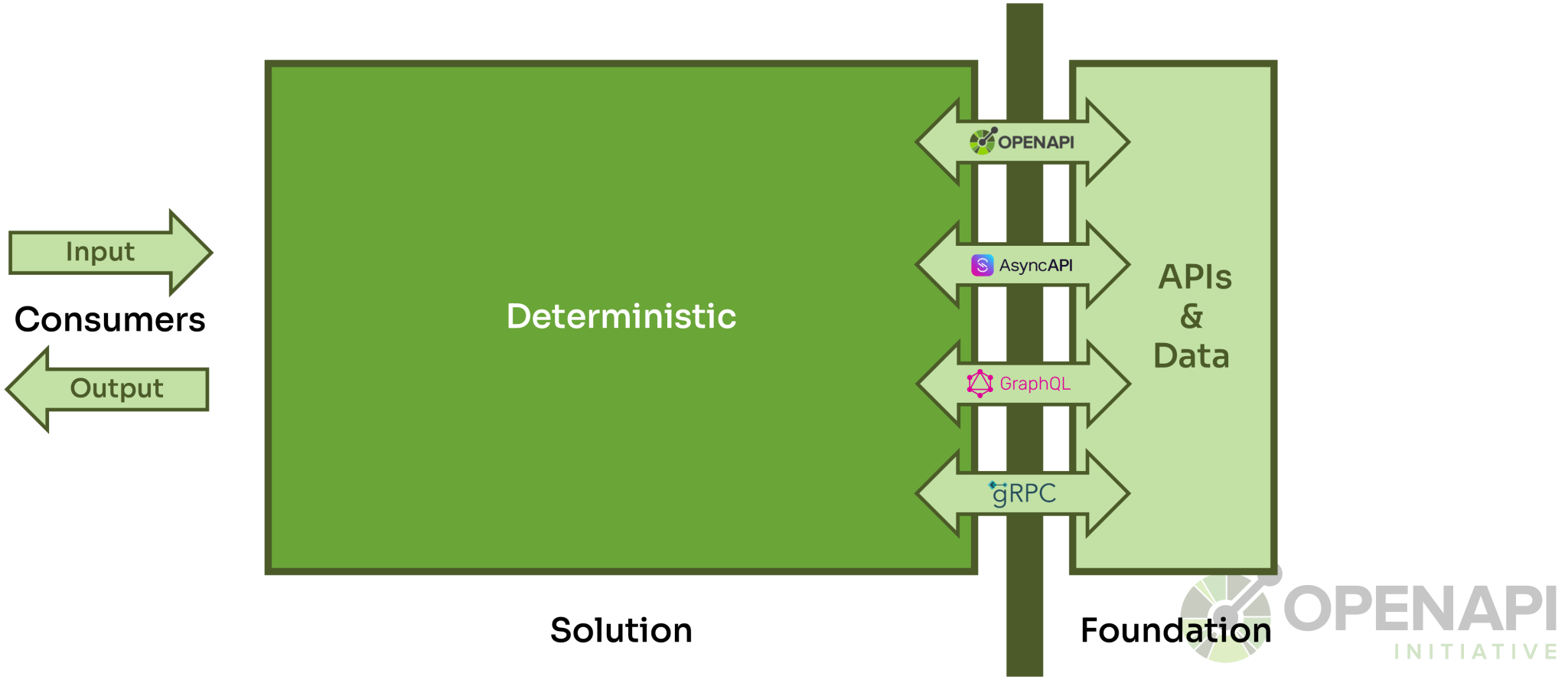
	CPU Computation	LLM Inference
Elapsed time	a few nanoseconds	~0.1–2 seconds
Computation	one integer division	~10^{11}–10^{13} FLOPs
Energy	~pJ–nJ	~1–100 J
Correctness	exact	not guaranteed

- CPU: One deterministic instruction
- LLM: Trillions of operations for probabilistic inference
- The LLM might still get the answer wrong

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Before AI

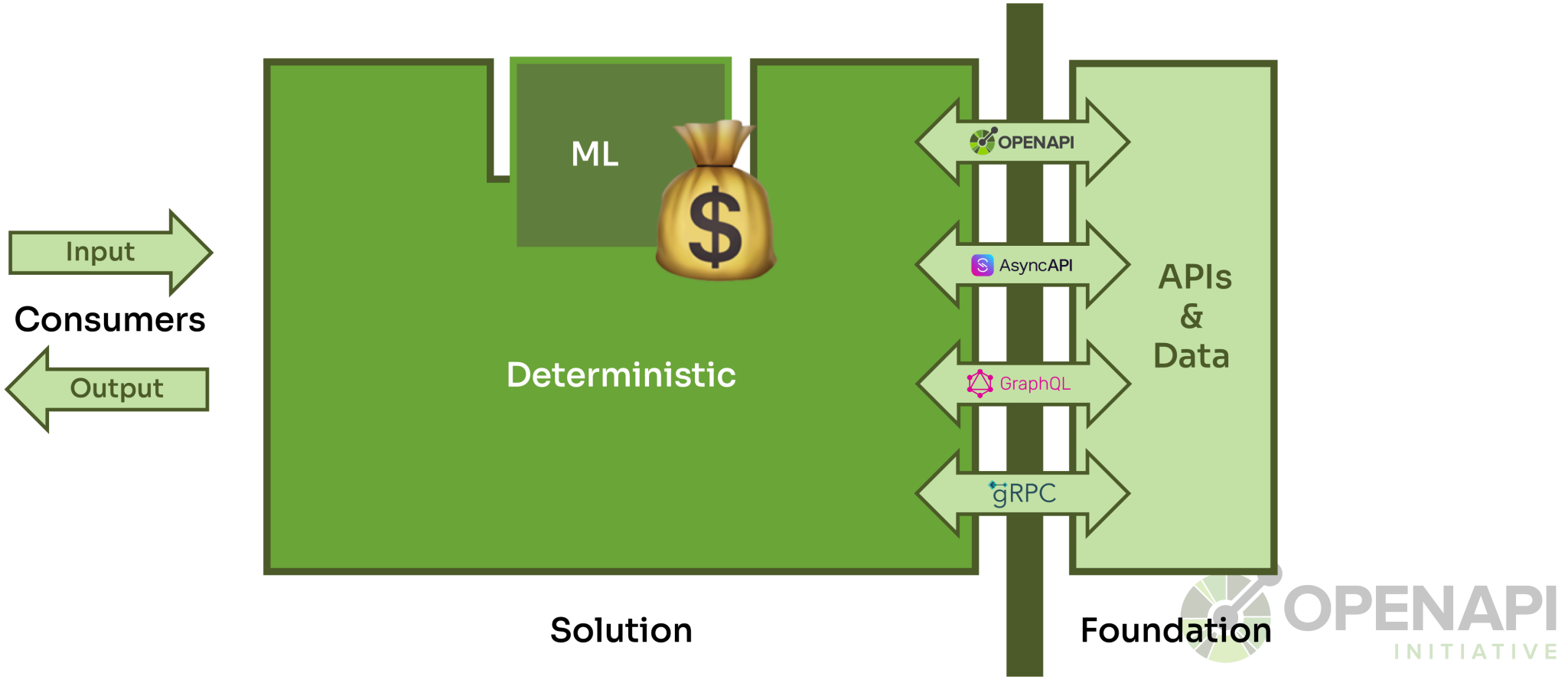
- Code runs deterministically



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Machine Learning (ML): 2010s

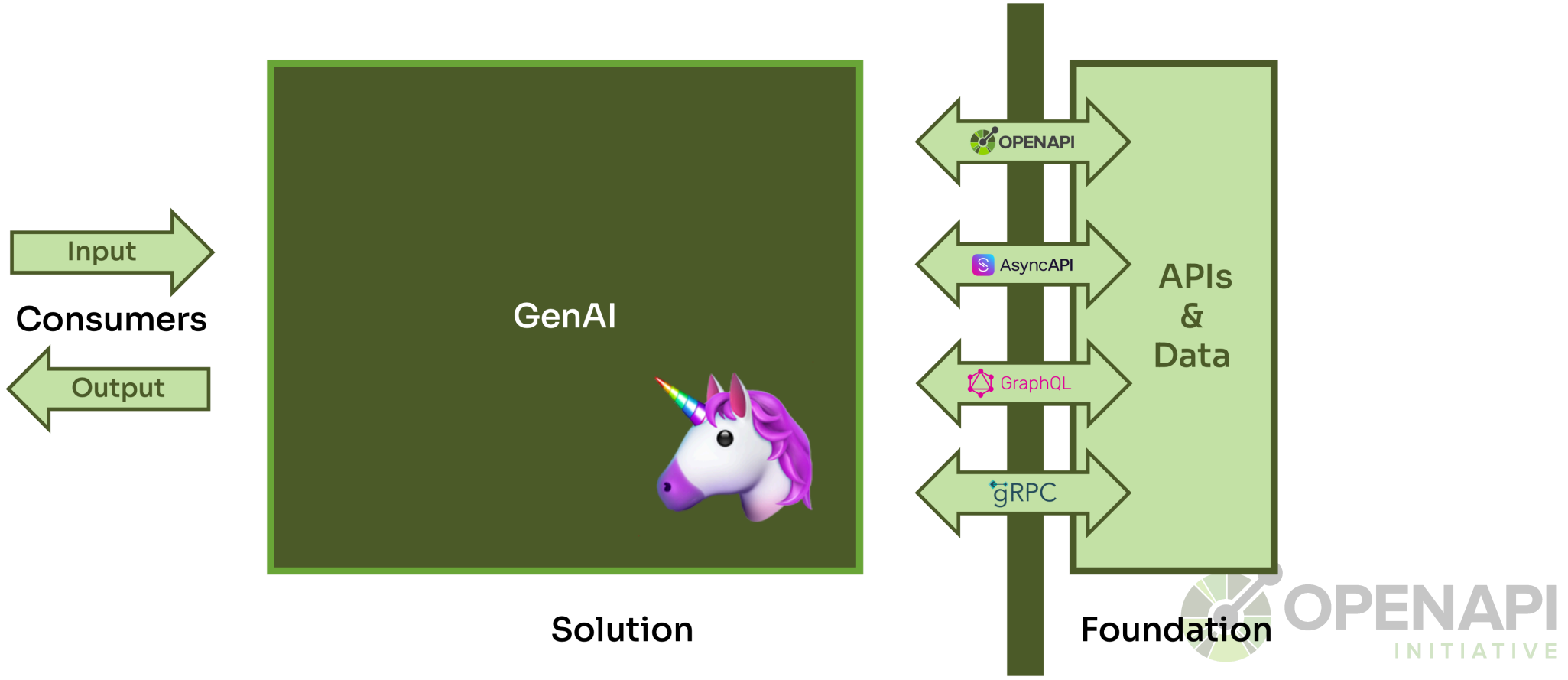
- Training models for specific problems becomes mainstream, but it is costly



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GenAI: 2022

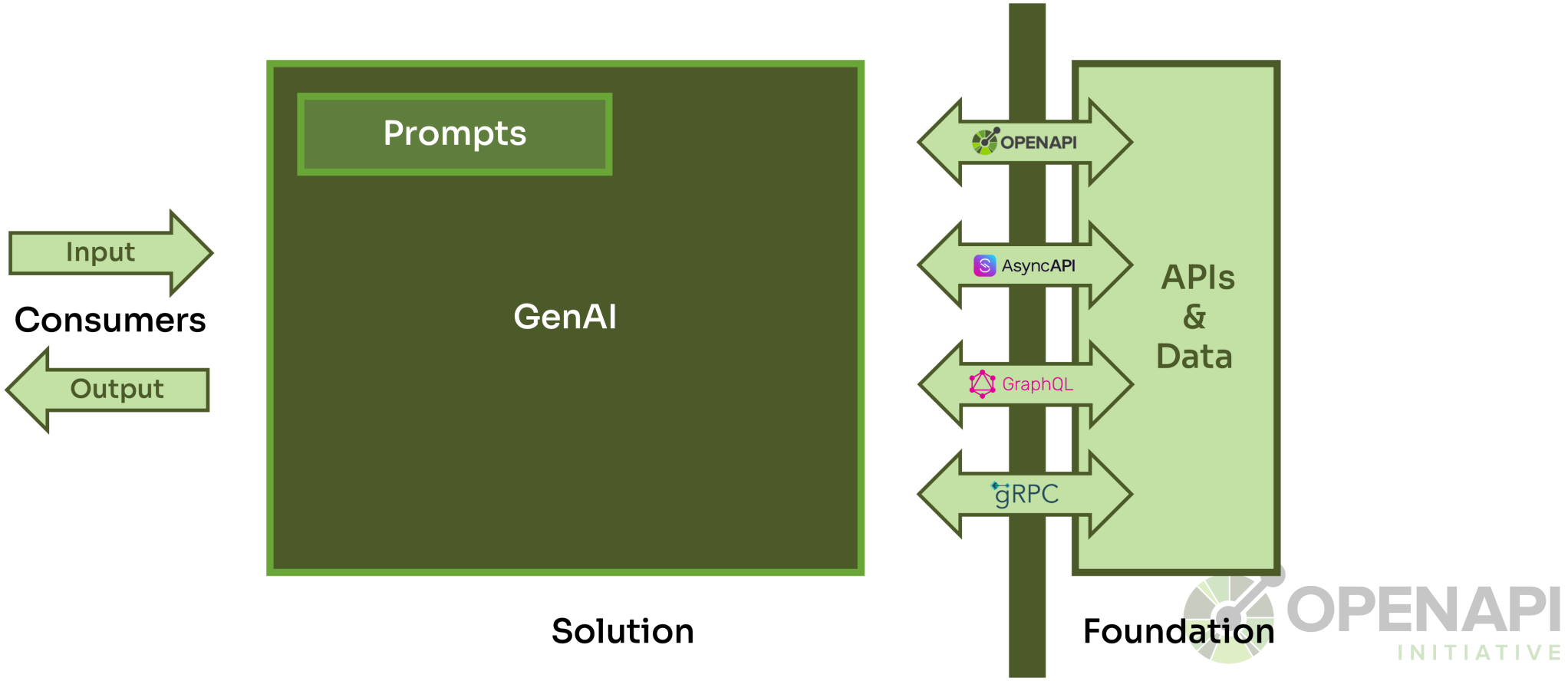
- LLMs make ML more widely applicable and much cheaper, but they are isolated



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Prompt Engineering: 2023

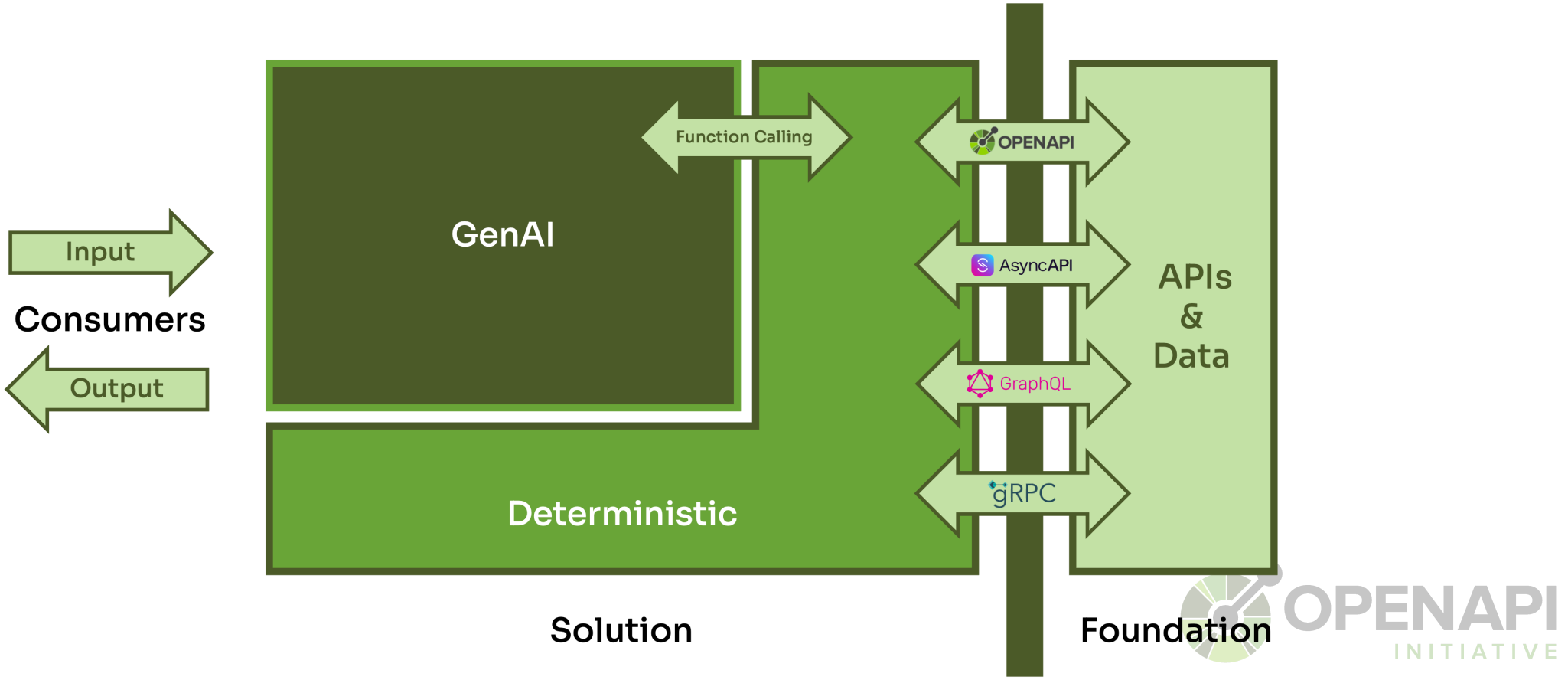
- First attempts to better manage probabilistic LLMs



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Function Calling: June 2023

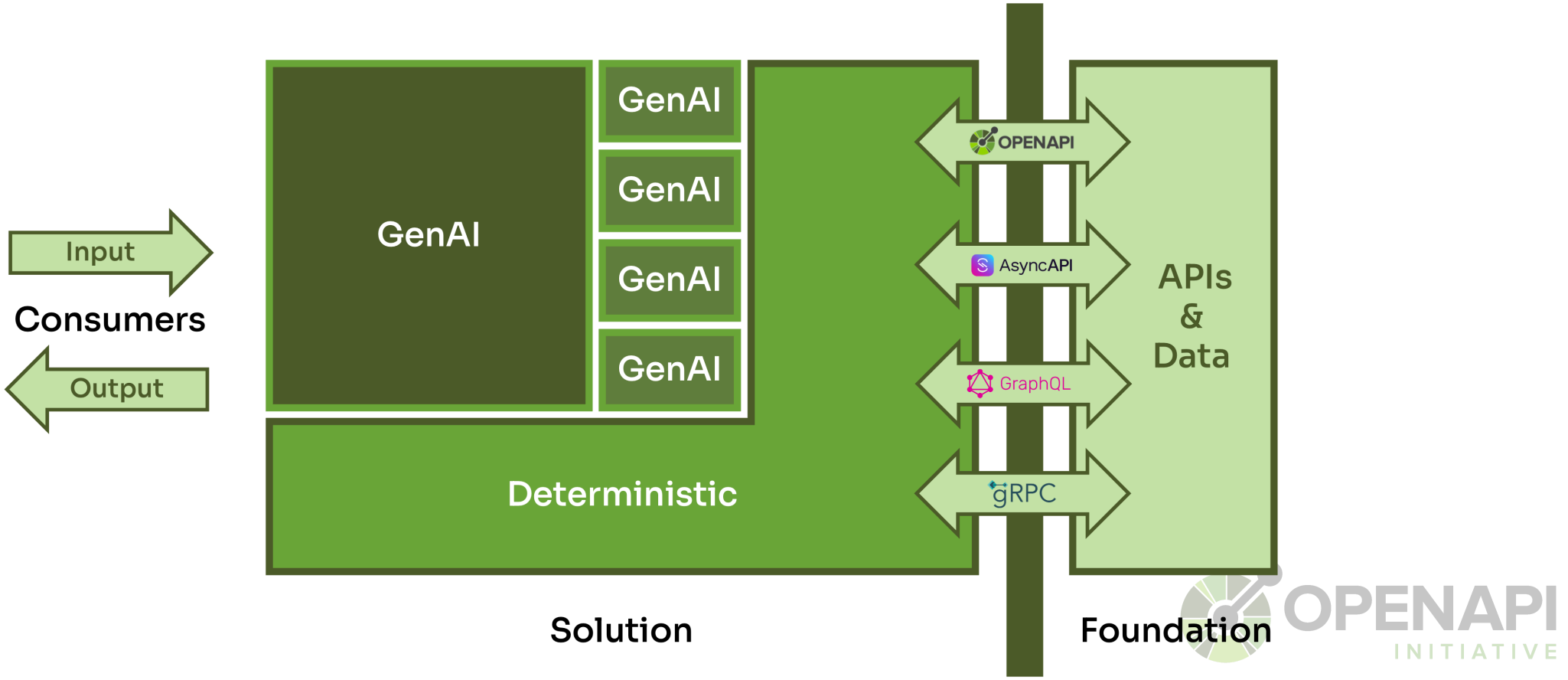
- LLMs are not just “a brain in a jar” anymore



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Subagents: 2024

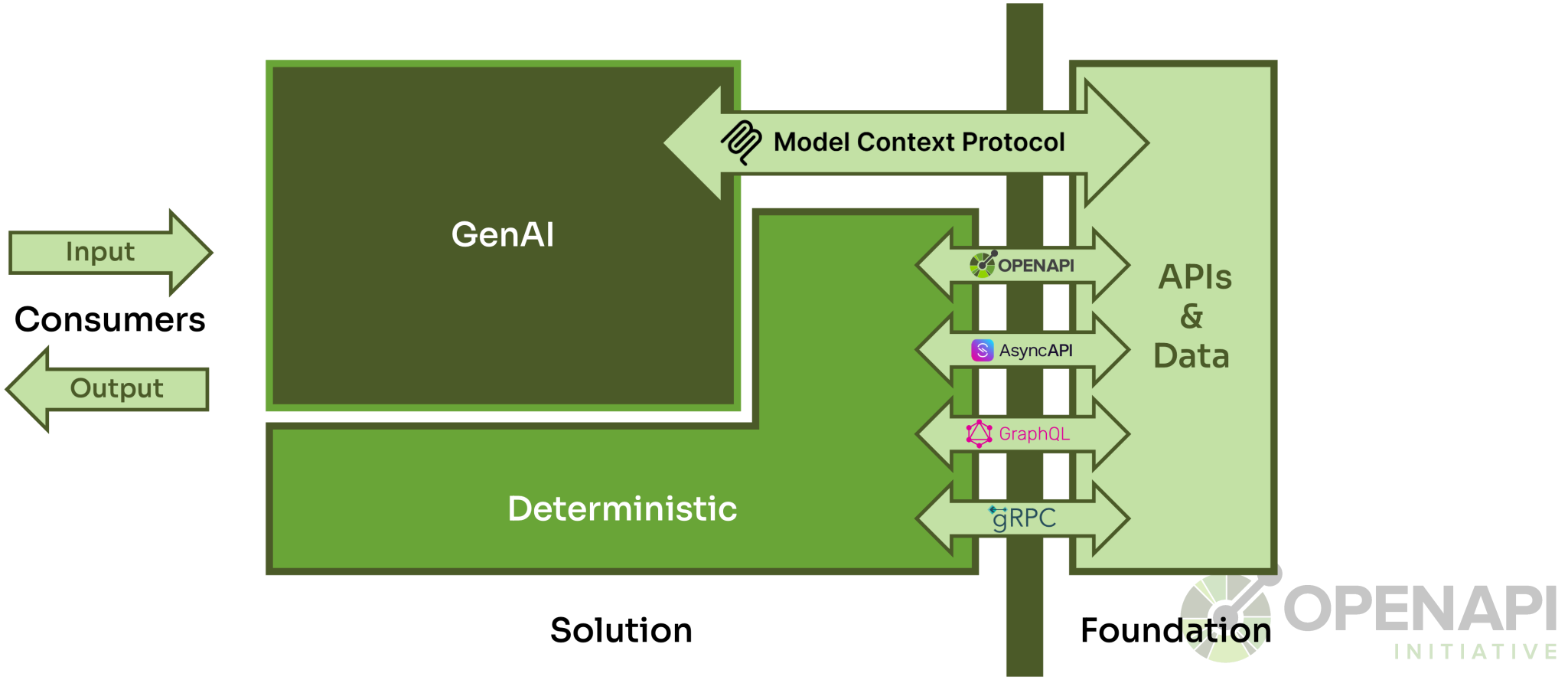
- Decomposing problems into sub-problems for specific contexts or models



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Model Context Protocol: November 2024

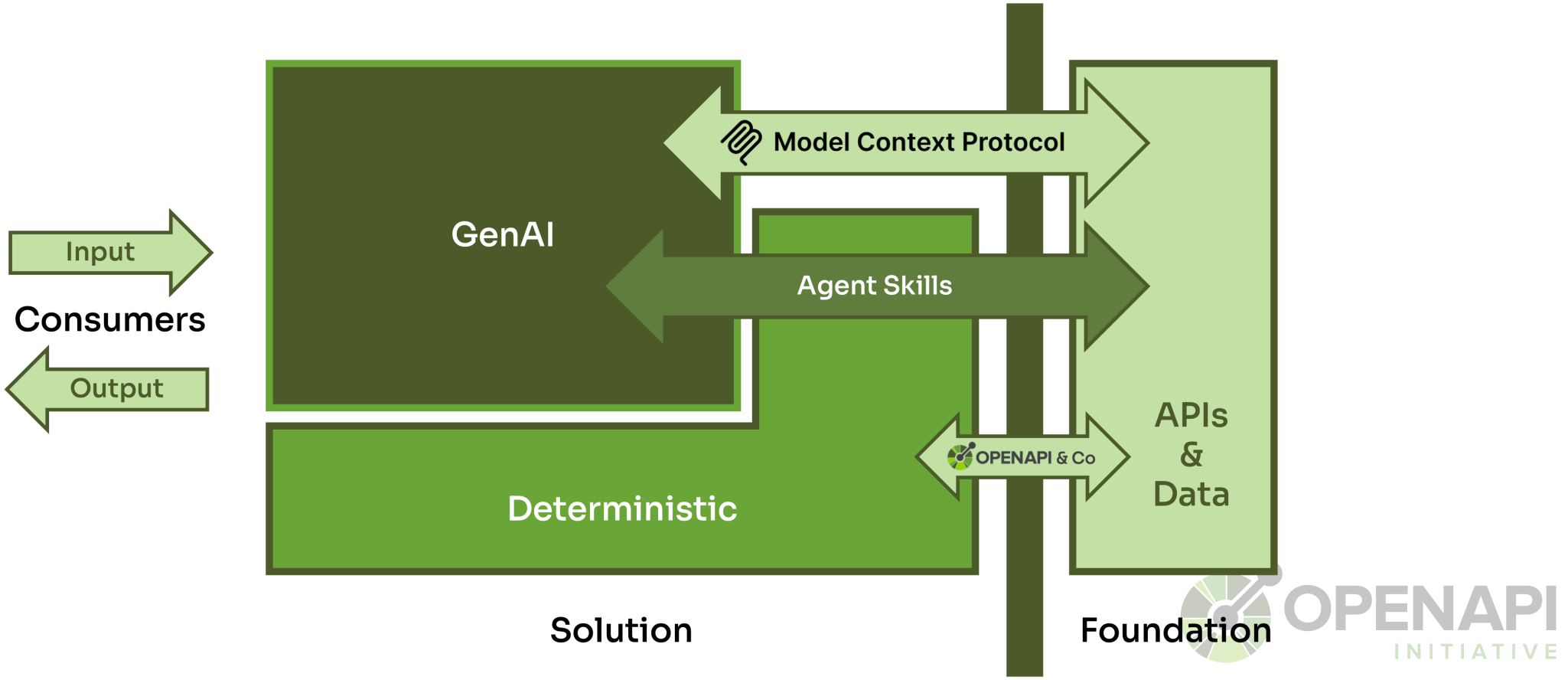
- Interact directly with context in a standardized way



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Agent Skills: October 2025

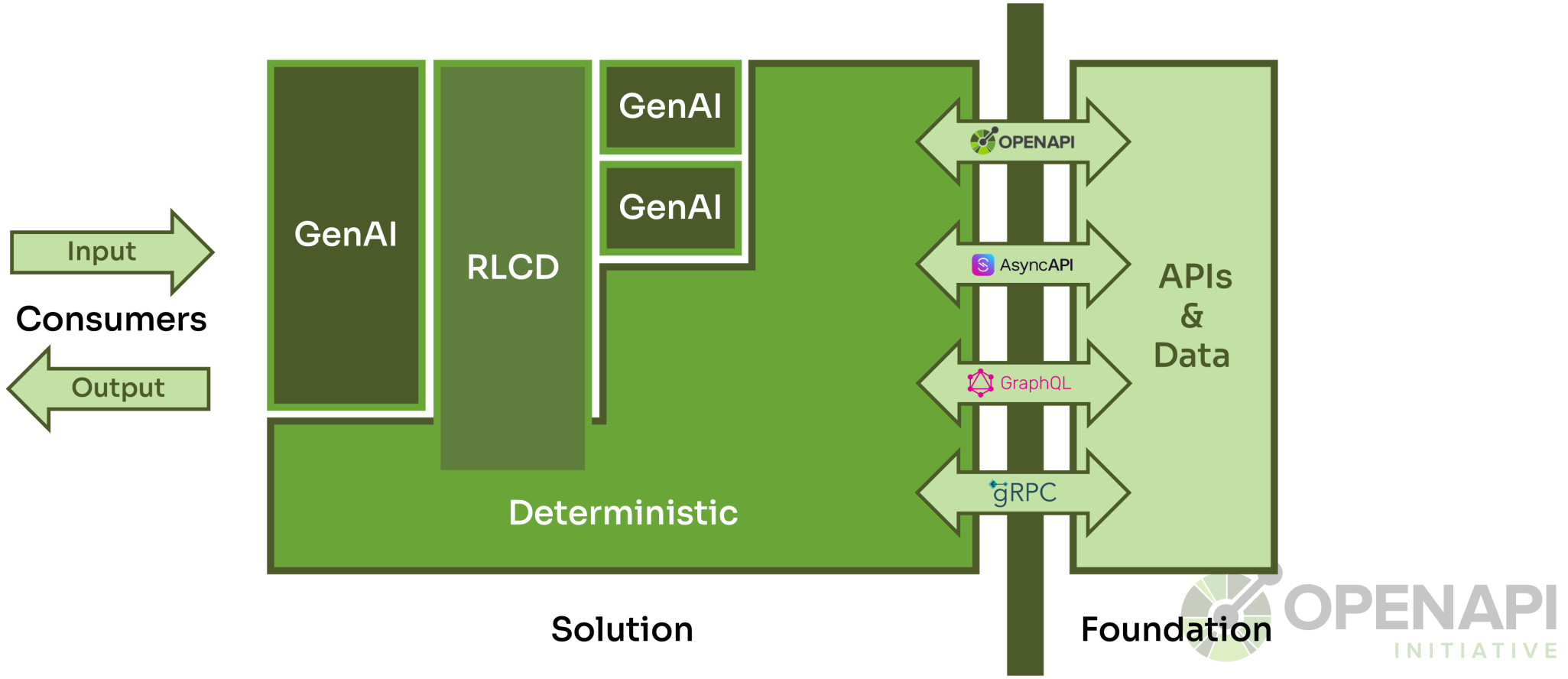
- Prepackaged mixes of prompts, local tools, and external tools



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Decision Models (RLCD): September 2026

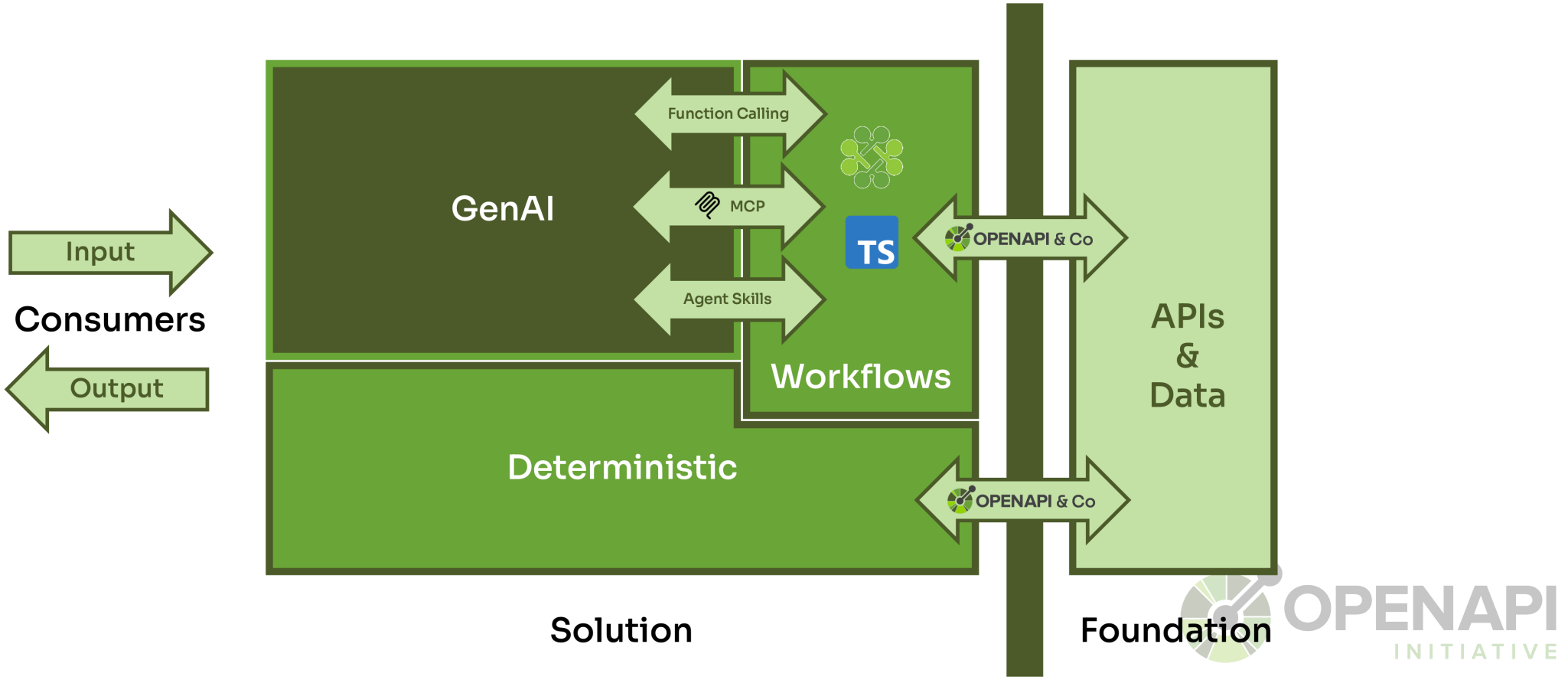
- Move from human-friendly LLMs to machine-friendly structured output



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Workflows

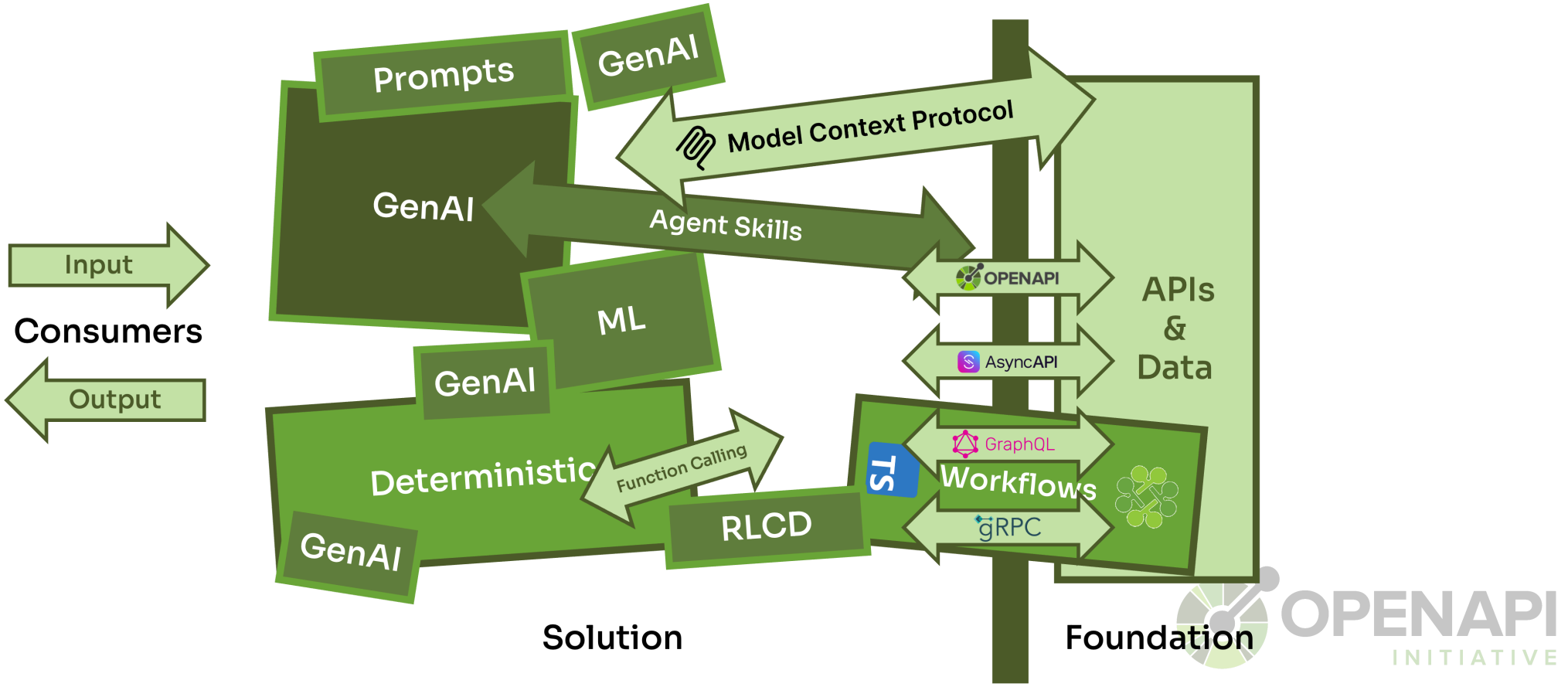
- Turn repeated processes into repeatable, reliable, reviewable, discoverable workflows



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Everything Everywhere All at Once

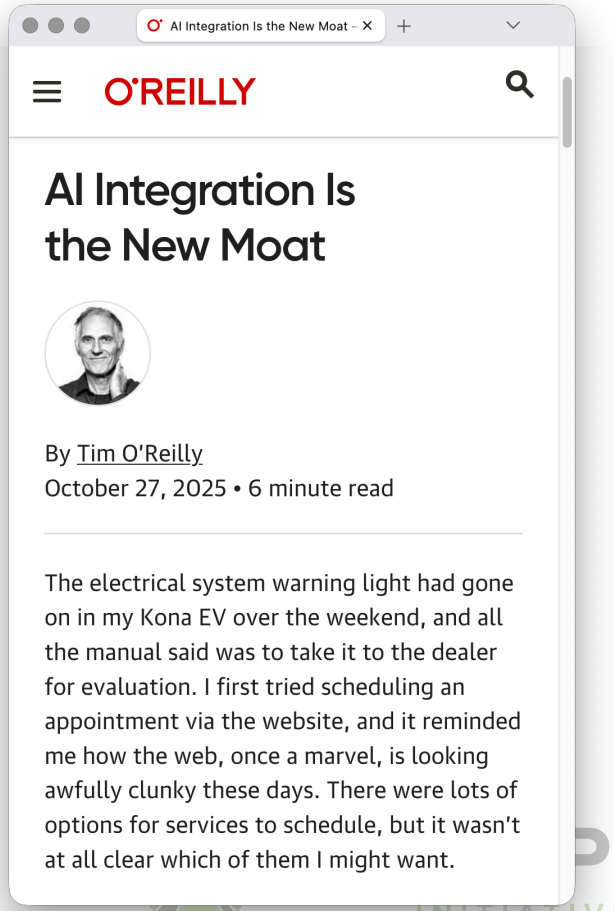
- Expect this menagerie to change at a rapid clip



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Solution Architecture Is Only Part of It

- All of these mechanisms are about composing a solution
- Every one of them has to interact with APIs and data
- Wiring that up is where many AI efforts stall or don't scale
- AI integration architecture matters as much as AI solution architecture



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Monday Tasks

- Keep watching what arrives, because the line will move again
- Modularize by subproblem, so that you can move the line later
- Invest in the APIs and data that everything else has to reach



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Thank You!

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